

Ex. No 5 Use Autopsy to create a case and import evidence

Description

Autopsy is an open-source digital forensics platform used for analyzing and extracting data from digital devices. Here's a step-by-step process on how to use Autopsy for a basic forensic investigation:

Drive link:

https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v

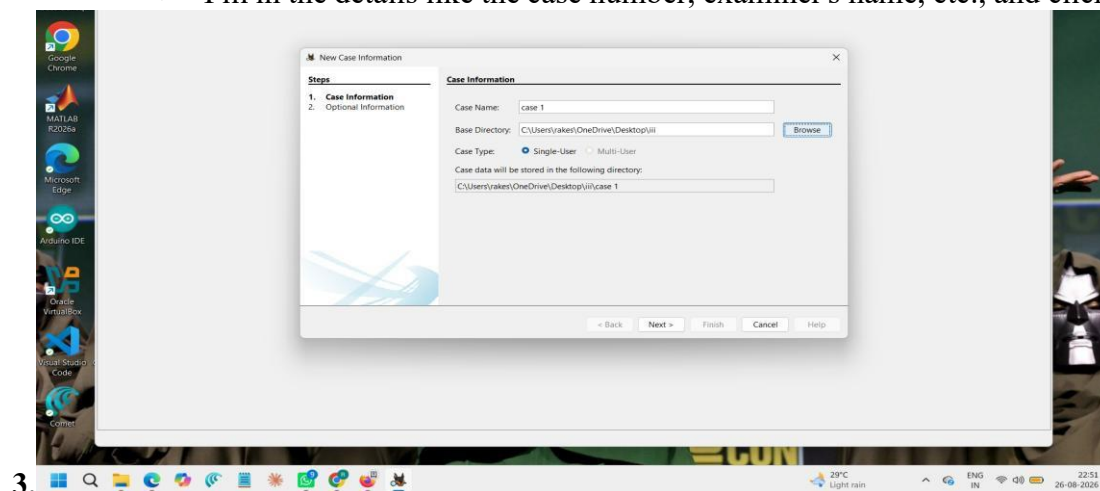
Download files: 4Dell Latitude CPi.E01, 4Dell Latitude CPi.E02

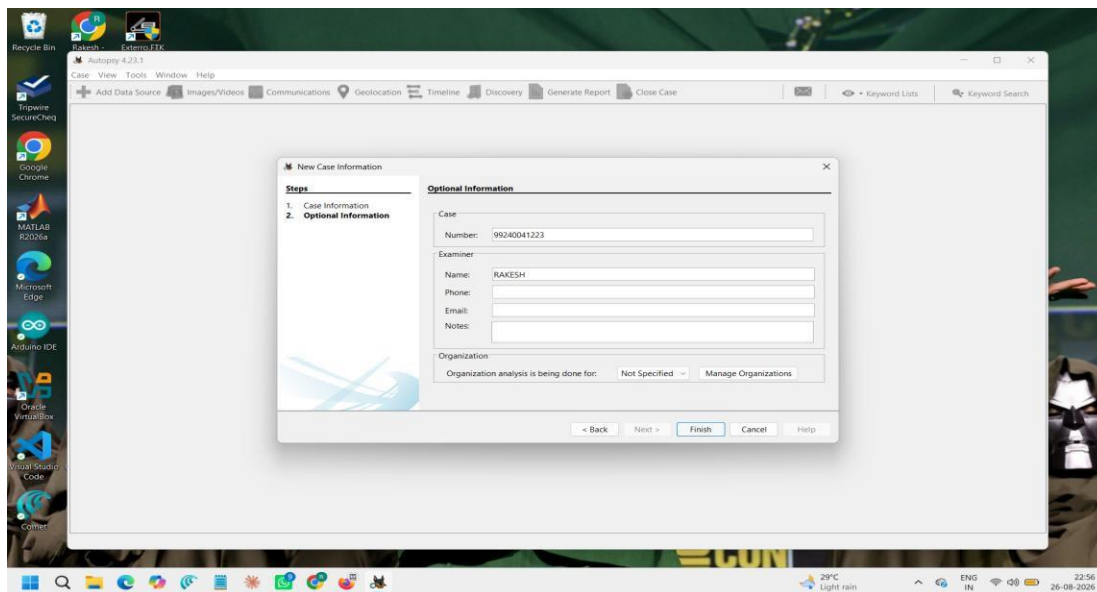
1. Installation

- **Download and Install:** Autopsy can be downloaded from the official website. Follow the installation instructions based on your operating system (Windows, Linux, or macOS).

2. Starting a New Case

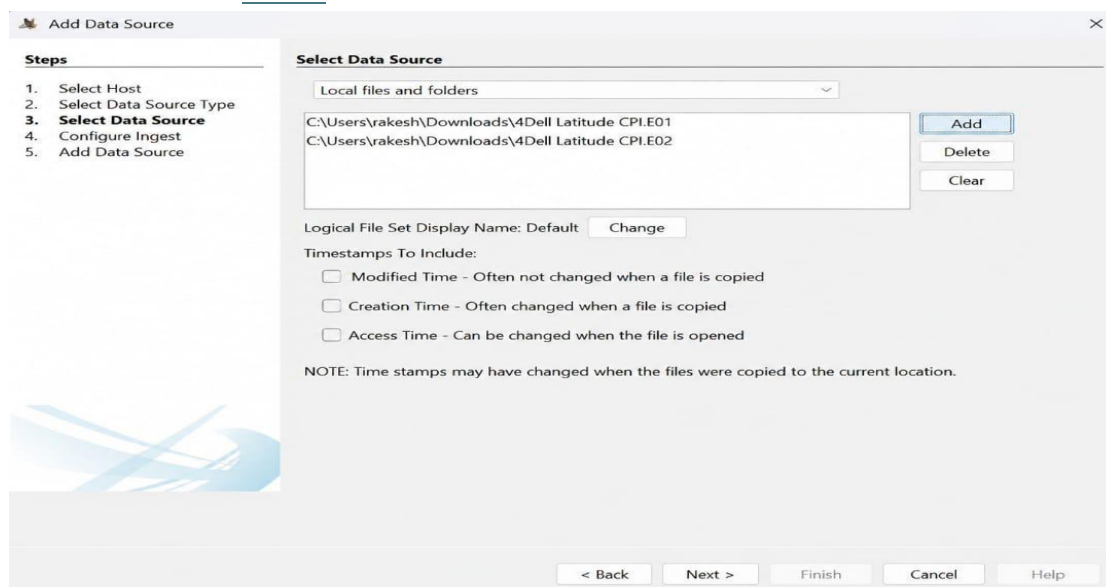
- **Open Autopsy:** Launch the application after installation.
- **Create a New Case:**
 - Click on New Case.
 - Enter the case name and location where the case data will be stored.
 - Fill in the details like the case number, examiner's name, etc., and click Next.





4. Adding a Data Source

- **Choose the Type of Data Source:**
 - After creating a case, you'll be prompted to add a data source.
- You can add disk images, directories, logical files, or local disks. **Select the Data Source:**
 - Browse to the location of the image file (e.g., .E01, .dd, or .raw), physical disk, or directory you want to analyze.
 - <https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixcxTU-v>



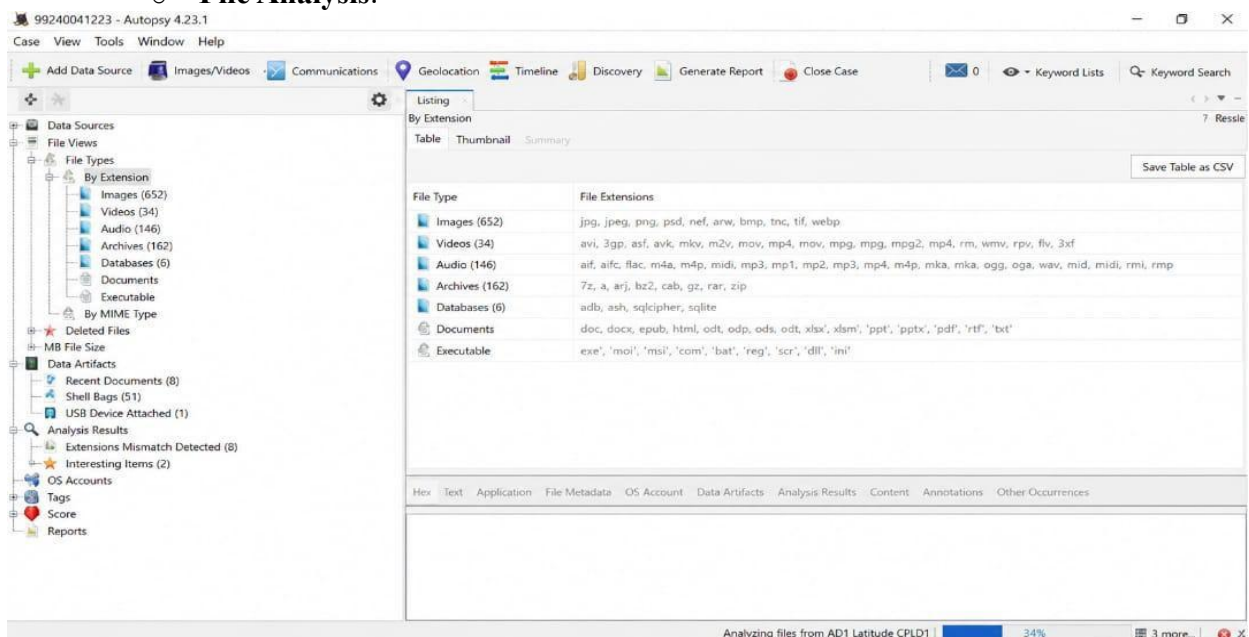
- **Configure Ingest Modules:**
 - Autopsy allows you to select specific analysis modules such as File Type Identification, Keyword Search, Hash Lookup, etc.
 - You can enable or disable these based on your investigation needs.
- **Start Analysis:** Click Next to start the analysis.

5. Initial Analysis and Overview

- **Ingest Progress:** As Autopsy processes the data source, you'll see the progress in the lower-left corner.
- **Explore the Resulting Artifacts:**
 - Autopsy automatically categorizes findings such as web artifacts, file system metadata, and communication records.
- **Use the Tree Viewer:**
 - On the left pane, you'll see a tree structure where you can explore different aspects like File System, Web History, Email, etc.

6. Detailed Analysis

- **Keyword Search:**
 - You can perform specific keyword searches using the Keyword Search module.
 - Use pre-configured lists or enter custom keywords.
- **File Analysis:**

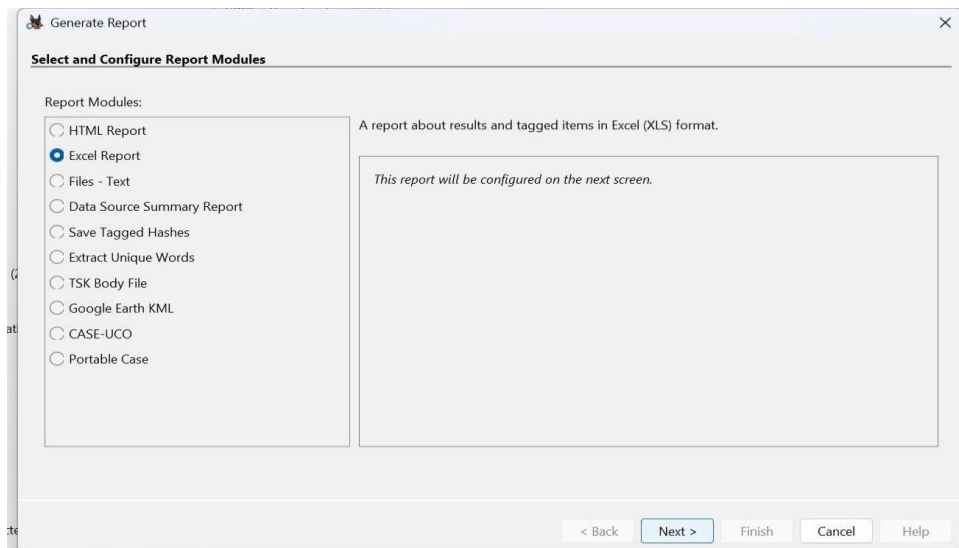


- Navigate through files and folders under the File Types or File System section.
- Open, view, or export files for further examination.
- **Timeline Analysis:**
 - Use the Timeline module to visualize events based on timestamps.
 - This can help track user activity over time.
- **Hash Analysis:**

- Compare file hashes with known databases to identify known good or bad files.

7. Reporting

- **Generate a Report:**
 - After analyzing the data, click on Generate Report from the toolbar.
 - Choose the type of report (HTML, CSV, Excel, etc.).
 - Select which parts of the analysis you want to include in the report.
- **Export Findings:**
 - Export individual files or artifacts that you need for your report or further analysis.
- **Final Review:**
 - Review the report to ensure it includes all relevant information.
 - Save or print the report for use in your case.

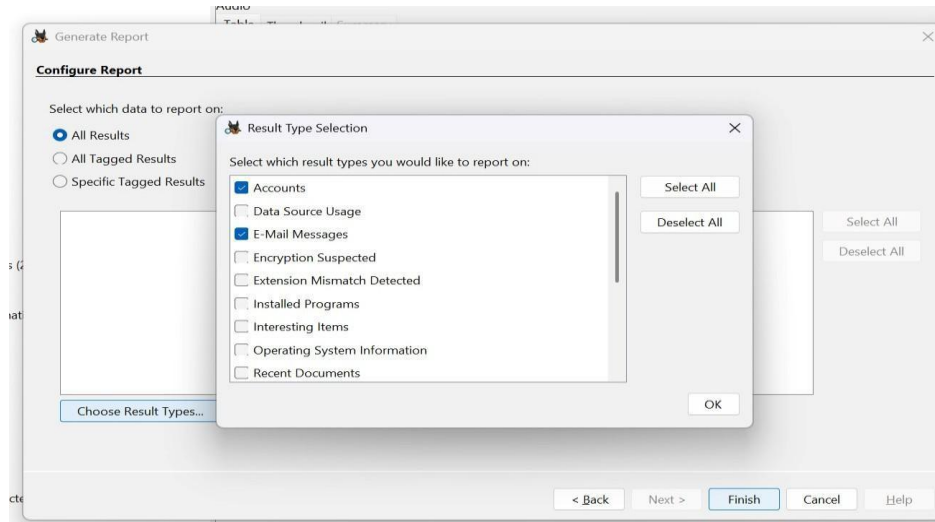


8. Case Closure

- **Close the Case:**
 - Once you have completed your investigation, close the case within Autopsy.
- **Archiving:**
 - Ensure all data and reports are properly archived according to your organization's policies.

9. Advanced Features (Optional)

- Autopsy allows for custom modules to be added if you need specific analysis tools not included by default.
- **Collaboration:**
 - Autopsy can be configured for multi-user cases if you're working in a team environment.s



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result: Case created in Autopsy, disk images imported and analyzed for artifacts. Report generated with findings on file system, web history, and communication records.

